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S2

Started on	Thursday, 24 September 2026, 9:30 AM
State	Finished
Completed on	Thursday, 1 October 2026, 9:28 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  // Function to find the duplicate element using Floyd's Cycle Detection
5  int findDuplicate(const int* arr, int n) {
6      // Phase 1: Finding the intersection point in the cycle
7      int tortoise = arr[0];
8      int hare = arr[0];
9
10     do {
11         tortoise = arr[tortoise];      // Moves 1 step
12         hare = arr[arr[hare]];        // Moves 2 steps
13     } while (tortoise != hare);
14
15     // Phase 2: Finding the entrance to the cycle (the duplicate number)
16     tortoise = arr[0];
17     while (tortoise != hare) {
18         tortoise = arr[tortoise];      // Both move 1 step
19         hare = arr[hare];
20     }
21
22     return tortoise;
23 }
24
25 int main() {
26     int n;
27
28     // Read the number of elements
29     if (scanf("%d", &n) != 1) {
30         return 1;
31     }
32
33     // Allocate memory for the array
34     int* arr = (int*)malloc(n * sizeof(int));
35     if (arr == NULL) {
36         return 1;
37     }
38
39     // Read the array elements
40     for (int i = 0; i < n; i++) {
41         if (scanf("%d", &arr[i]) != 1) {
42             free(arr);
43             return 1;
44         }
45     }
46
47     // Find and print the duplicate
48     int duplicate = findDuplicate(arr, n);
49     printf("%d\n", duplicate);
50
51     // Clean up memory
52     free(arr);

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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